

detect the links; in fact, it failed to pick up the small-sized fonts.

Combi document: This was a test of the OCR package's ability to read special symbols (such as currency symbols, Greek alphabets and mathematical notations) commonly found in business and scientific documents. The formatting on this document included bold and italicised text and two different sets of reverse fonts—black text on grey background and white text on grey background.

Overall, the results here were quite disappointing for accuracy as well as retaining formatting. All five software found it particularly difficult to detect the black text on grey background.

The best performer here was ABBYY FineReader 6.0 CE, with an accuracy of 96.77 per cent. It was also the best when it came to retaining formatting—it scored 2.5 out of 5. However, like the other software, it was unable to detect Greek symbols such as 'µ' and 'Ø' and the currency symbols. Of the nine currency symbols included in the document, the software were only able to read \$, £ and ¥—PageGenie Pro 4.0 simply ignored the entire line containing currency symbols and returned junk characters for the white text on grey background.

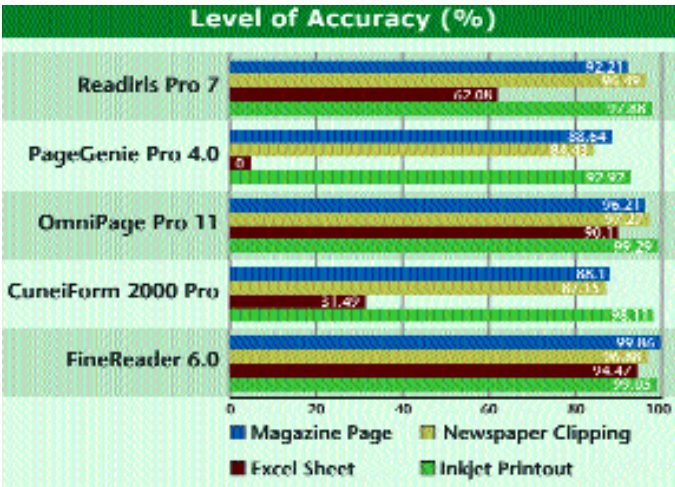
CuneiForm 2000 logged only 69 per cent accuracy—the poorest results here. The document was full of junk characters and reverse fonts were missing entirely.

Excel spreadsheet: If you have a large number of bills, *chalars* or complex Excel spreadsheets then the results of this test should be of interest to you. Here the ability of the OCR software to read characters accurately as well as retain formatting are equally important.

Our document had a table with native Excel formatting such as horizontally and vertically merged cells, text writ-

Motherboard Test Score Log for May 2002		
Brand: DFI	Model: DFTX8150N	Revision: 1.00 (V/N)
Manufacturer: CPU: Pentium 4	Processor: Intel Celeron	Processor: Intel Celeron (V/N)
Memory: 512MB	Memory: 512MB	Memory: 512MB (V/N)
Hard Drive: 40GB	Hard Drive: 40GB	Hard Drive: 40GB (V/N)
Optical Drive: DVD-ROM	Optical Drive: DVD-ROM	Optical Drive: DVD-ROM (V/N)
Sound Card: Creative	Sound Card: Creative	Sound Card: Creative (V/N)
Network Card: Realtek	Network Card: Realtek	Network Card: Realtek (V/N)
Modem: 56K	Modem: 56K	Modem: 56K (V/N)
Power Supply: 300W	Power Supply: 300W	Power Supply: 300W (V/N)
Case: Mid Tower	Case: Mid Tower	Case: Mid Tower (V/N)
OS: Windows XP	OS: Windows XP	OS: Windows XP (V/N)
BIOS: Award	BIOS: Award	BIOS: Award (V/N)
Test Date: 05/05/2002	Test Date: 05/05/2002	Test Date: 05/05/2002 (V/N)
Tested By: [Name]	Tested By: [Name]	Tested By: [Name] (V/N)
Score: 100%	Score: 100%	Score: 100% (V/N)
Comments: All tests passed	Comments: All tests passed	Comments: All tests passed (V/N)

The Excel spreadsheet



ten in vertical format, and coloured cells with text and data. We also checked to see if the vertical text and ® symbol were detected after saving the file in Excel.

ABBYY Finereader 6.0 CE and OmniPage Pro 11 both scored over 90 per cent in accuracy and also retained the layout. PageGenie Pro 4.0, on the other hand, did very poorly here—the entire document was filled with junk characters and was saved in one column only (it scored zero for both, layout and accuracy). CuneiForm 2000 and Readiris Pro 7 also scored nothing for layout and formatting here! Also, none of the OCR software in this test were able to retain the colour of the cells.

Dot matrix printout: Dot matrix printouts really put the OCR engine to test—here the characters are formed using fine dots and it is a great challenge for the OCR engine to recognise it as a complete letter.

A B B Y Y
FineReader 6.0 CE again displayed its prowess in recognising characters and returned an impressive 95.28 per cent accuracy. It did miss out some characters, reading '0' as 'Q' and 'Z' as '2'. OmniPage Pro 11 came in second with a decent 91.03 per cent accuracy. Once again, only ABBYY

FineReader 6.0 CE and OmniPage Pro 11 retained the layout to a certain degree and scored 2.49 out of 5 each.

The poorest performer here was CuneiForm 2000; it scored a paltry 23.82 per cent in the accuracy test and had lots of misspelled words and junk characters. It was also the worst of the lot at retaining formatting—not a single formatting element was retained and the entire document was a mess.

Photocopied document: A photocopied document tests the true mettle of the OCR engine—the text is usually faded and there are smudges on the sheet. For this test, an inkjet printout was photocopied at normal quality mode.

ABBYY FineReader 6.0 CE once again came out tops, with an accuracy level of 95.75 per cent—there were no junk characters in the OCR-ed document! But it scored only 3.32 out of 5 for formatting.

Tips for Better OCR Results

- A document which is free of wrinkles, tears, smudges or creases will yield more accurate results.
- Keep a sheet of white paper behind the document when you scan it for OCR-ing, especially when scanning newspapers and thin paper. This will help eliminate smudges during scanning.
- Use the Rotate and Crop tools of an imaging software to align the text if there is no Deskew option in your OCR software.
- Despeckle the scanned image before you begin OCR-ing the document. This will remove the dots (background noise) and help the OCR engine recognise neighbouring fonts correctly.
- Scan the image in full colour mode. If the document has simple black fonts on a white background, use the greyscale mode. You will get poor results if you select Line art (black and white) mode.
- Use the image splitting tool to manually define the text and image area. This will help the OCR engine retain the format and layout more accurately.
- If your documents have custom fonts and formats that are not picked up by the OCR software, train the OCR software by adding those specific fonts and letter types into its dictionary.